

SECTION B*Answer all questions in the spaces provided.*

9. (a) Consider the elements labelled **A-G**. These are not chemical symbols.

A	$1s^2 2s^2 2p^1$
B	$1s^2 2s^2 2p^3$
C	$1s^2 2s^2 2p^6$
D	$1s^2 2s^2 2p^6 3s^2$
E	$1s^2 2s^2 2p^6 3s^2 3p^1$
F	$1s^2 2s^2 2p^6 3s^2 3p^6$
G	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

- (i) Give the letter (**A-G**) of the element with the largest **first** ionisation energy.

Give **two** reasons for your answer.

[3]

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- (ii) Give the letter (**A-G**) of the element with the largest **last** ionisation energy.

Give a reason for your answer.

[2]

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- (iii) Give the letters (**A-G**) of **all** the elements above that are metals.

[2]

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- (iv) Give the letters (**A-G**) of **all** the elements above that form basic oxides.

[1]

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(b) Magnesium exists as three naturally-occurring stable isotopes. They can be identified using a mass spectrometer.

(i) State how magnesium ions are formed in a mass spectrometer. [1]

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(ii) State how magnesium ions are separated in a mass spectrometer. [1]

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(c) Magnesium-28 is an unstable radioactive isotope that decays by β -emission.

(i) Give the mass number and symbol of the element formed as a product of the radioactive decay of magnesium-28. [1]

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(ii) If it takes 84 hours for the activity of the isotope to decay to $\frac{1}{16}$ th of its original activity, calculate its half-life. [1]

Half-life = hours



- (d) Magnesium is a typical metal. Describe the bonding in magnesium and explain why it is ductile. You may include a diagram as part of your answer. [3]

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- (e) According to the label on the bottle, the concentration of magnesium ions in a sample of Welsh mineral water is 15 mg / litre.

Calculate the concentration of magnesium ions in mol dm^{-3} . [1]

Concentration = mol dm^{-3}

